Hardware Diagnostic & Infrastructure Report

NCB TECHNICAL SOLUTION — PROFESSIONAL ARCHITECTURE & SERVICE

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Location: Harihar, Karnataka, India
Status: Technical Deployment Finalized

1. Physical Device Verification

📷 **Figure 1: Internal Chassis Components**

Physical hardware verification log from Harihar, Karnataka. Confirms factory 1 TB SATA HDD layout, cooling fan module, memory placement, and system board architecture.

📷 **Figure 2: Regulatory Identity Label**

Lower panel regulatory sticker scan verifying system attributes: Lenovo IdeaPad 3 15IML05, MTM 81WB01D7IN, Input 20V/2.25A, and Serial Number PF392NA2.

2. Device Identity & Specifications

System Metric	Verified Technical Parameters
Brand & Series	Lenovo IdeaPad 3
Full Model Name	IdeaPad 3 15IML05
Model Number (MTM)	81WB01D7IN
Serial Number (SN)	PF392NA2
Machine Order (MO)	PF9XB1929147
Manufacture Date	September 29, 2021
Processor (CPU)	Intel Core i3-10110U (10th Generation) • 2 Cores | 4 Threads • 2.1 GHz Base | 4.1 GHz Turbo
Current Memory (RAM)	8 GB DDR4 2666 MHz • 4 GB soldered onboard + 4 GB removable SO-DIMM module
Factory Storage	1 TB SATA HDD (5400 RPM)
Display Panel	15.6 inch Full HD (1920x1080) Anti-Glare
Operating System	Windows 11 Home (64-bit)

3. Expansion Capability Discovery

System Architecture Board Inspection Findings:

Physical diagnostic imaging of the internal motherboard confirms additional high-speed interface availability alongside existing storage infrastructure.

Storage Interface Slot	Status	Engineering Recommendation
SATA Interface	✓ OCCUPIED	Retain existing 1 TB SATA HDD purely for secondary high-capacity data archives.
M.2 NVMe Slot	✓ AVAILABLE (OPEN)	Install a **256 GB NVMe SSD** or **512 GB NVMe SSD**. Migrate the Windows 11 installation onto this solid-state drive.

4. Peripheral Interface & Connectivity Layout

Interface Type	Integrated Onboard Port Specifications
USB Connectivity	2x USB 3.1 Gen 1 (Type-A) | 1x USB-C (USB 3.1 Gen 1)
Video Interface	1x HDMI 1.4b terminal
Media Reader	4-in-1 SD Card Reader
Audio & Camera	3.5mm Headphone / Microphone Combo | 720p HD Camera with Privacy Shutter Array
Acoustic & Wireless	2x 1.5W Speakers with Dolby Audio | 802.11ac Wi-Fi + Bluetooth Integration

5. Diagnostic Findings & Root Cause Analysis

Symptom Log: Extreme System Lagging, Freezing, and Sudden Black Screen Drops

Resource utilization logging isolates performance degradation to structural storage and configuration strain:

Analysis Vector	Technical Mechanism	Operational Impact
Mechanical Disk Scurrying	Windows 11 background updates and active indexing frequently draw continuous small data packets that overload a standard mechanical drive.	Sustained **100% active disk usage** causing immediate system freezes.
Resource Paging Constraints	When application memory requirements exceed optimal system thresholds, data transactions spill into drive caching spaces.	The 5400 RPM drive queues fail, causing display and system execution time-outs.

6. Physical Maintenance Operations Executed

The following targeted internal physical service tasks were performed directly on the unit:

- **Thermal Service:** Battery connector cleanly disconnected before performing internal maintenance. Removed accumulated dust build-up from the cooling fan blades and the matching heatsink channels to restore factory airflow parameters.
- **RAM Maintenance:** Removable RAM module safely unseated and carefully inspected. All contact surfaces were thoroughly cleaned to ensure clean continuity, and the module was reseated securely back into its SO-DIMM expansion slot.

7. Recommended Upgrade & Performance Optimization Plan

Recommended Step	Implementation Mechanics	Expected Performance Lift
M.2 NVMe SSD Integration	Install a dedicated 256 GB or 512 GB high-speed solid-state drive into the discovered open M.2 terminal. Fresh install or clone Windows 11 to run strictly from the SSD.	Eliminates operating system dependency on slow mechanical storage paths.
RAM Configuration Upgrade	Remove the existing 4 GB removable module from the SO-DIMM slot and replace it with an 8 GB DDR4 module. *(Resulting in 4 GB Onboard + 8 GB Module = 12 GB Total RAM)*	Provides necessary memory headroom for modern multitasking and eliminates memory paging drops.
Repurpose 1 TB SATA HDD	Retain the 1 TB factory drive inside its existing bay, wiping system files and utilizing it purely as a secondary cold-storage option for mass media and archives.	Retains massive storage space without impacting active computing speeds.

8. Post-Upgrade OS Integrity & Security Verification

Following full physical setup, physical hardware component upgrades, and completion of all core Windows system updates, the following deployment optimization and deep architecture checks were executed to finalize system baseline parameters:

Step A: Advanced System Image Repair (DISM)

Access: `Command Prompt (CMD)` **Privilege:** **Run as Administrator**

Utilizes the Deployment Image Servicing and Management (DISM) engine to verify the baseline health of the local Windows OS image component store. Run this if SFC fails.

```
DISM /Online /Cleanup-Image /CheckHealth
```

DEEP REPAIR To actively pull clean files down via Windows Update to overwrite corrupted OS components, run:

```
DISM /Online /Cleanup-Image /RestoreHealth
```

Step B: Built-in Windows Malware Scanning

Shortcut: `Win + R` **Target:** Run Dialog

Triggers the standalone Microsoft Malicious Software Removal Tool (MRT), allowing deep scanning of system directories independent of standard active antivirus agents.

```
mrt
```

9. Service Valuation & Billing Summary

Diagnostic & Maintenance Service Item	Amount (INR)
Standard Hardware Diagnostics & Fault Architecture Profiling	₹350.00
Internal Infrastructure De-dusting & Thermal Airflow Service	₹450.00
SO-DIMM RAM Module Interface Inspection, Cleaning, & Reseating	INCLUDED
Post-Upgrade OS Deployment & Administrative Integrity Scripting (DISM/MRT)	INCLUDED
Total Physical Service Charge (Labor)	**₹800.00**

Note: Component upgrade hardware pricing (NVMe SSD & 8GB DDR4 RAM Module) is pending client selection and will be invoiced separately upon part procurement.

10. Technical Review Summary & Outlook

Engineering Conclusion: Physical thermal servicing and RAM cleaning have been successfully completed. Following the implementation of the dual-channel 12 GB RAM layout change and primary storage migration to an NVMe solid-state module, the updates are **expected to significantly improve system responsiveness and reduce black-screen and freezing issues.**